Kapasa Makasa University

Information Communication Technology Department
(Cyber Security)

Wireless Security

(CYS331)

Lecture 2

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Wireless data networks

introduction

We can say that wireless networks are those which are not connected by any type of cables and it uses different wireless connections between network nodes. It is a process by which different home installations, companies installations and other installations on large scale has become easy because it avoid the costly and difficult process of introducing cables into the homes and offices areas because there is a complete process of connecting different devices on different location.

BACKGROUND

The first wireless network was developed under the brand **ALOHA** at **university of Hawaii** between1969 to 1970 but it was officially functional in June 1971, the first commercial wireless network was the **WAVELAN (brand name or family name of wireless network technology)** product family developed by **NCR** in 1986.

Need of wireless data networks

- Wireless network enables many devices to connect with each other and to connect with the same network without any data cables.
- They also help the users to share the data, files and resources with each other devices without any interruption.
- They allow every device such as smartphones, laptops, watches to move in the network area freely without any network distortion and loosing internet facility and still maintain the connection between devices and internet.

Benefits of wireless data networks

- Efficiency status: This help to improve the data transfer and communication in a faster way between multiple users just like Bluetooth which help to save energy and make communication easy.
- Access and range: Because of wireless networks we are able to use it freely without any cables and routers it allows us to communicate while we are moving to access different networks whether it is house network or office.
- Expenditure: it is easier and cost saving to install especially in buildings or where cables are not allowed in the places to install cables.
- Latest opportunities: now a days wireless networks are installed in airports lounges and train stations have hotspot facilities to allow users to connect their devices while they are travelling.

Disadvantages of wireless networks:

- security
- Installation criteria
- Coverage
- speed

Types of wireless data networks

- **Wireless local area network**
- **Wireless metropolitan area network**
- **Wireless personal area network**
- **Wireless wide area network**

WIRELESS LOCAL AREA NETWORK

- 1. This technology provides the internet access to the users inside the premises or in a limited outdoor area. It is not only used in homes and offices but also in stores and cafeterias.

- **In a local area network** a group of computers and other devices have a common wireless network within a distinct area. The architecture design of home networks are just as there is a modem connected to the cable or fiber cables provided by the local service provider. Then a wireless router helps the modem to receive the signals from the modem due to this process signals are transmitted using the wireless protocol.

The networks in offices are little bit more complicated because there were the access points which are adjusted on the ceilings of the building that helps in broadcasting a wireless signal to the surrounding area. So this is the reason multiple access points are required in the buildings connected to the office network via a wired connection to the switch.



Types of wireless LAN's:

1. Infrastructure

   Most of the Wi-Fi networks are function in infrastructure mode so that wireless clients such as laptops and other smart devices connect to the wireless access point to join the available network.

1. Peer to peer

   In peer to peer wireless devices directly communicate with each other within the range without involving any access points. For example two computers directly communicate with each other to form a network

1. Wireless distribution system

In wireless distribution system wireless is expanded using multiple access points without the need for wired backbone to link them

- **Wireless metropolitan area network:**

This network provide access to the people outside of the office or home. This network has been installed in cities. This cover a wide area than local area network. But less than the wireless wide area network.

In this access points are placed on the poles or on the buildings through the area. When users are connecting to their destination they are connected to the nearest access point.



Advantages of metropolitan area network:

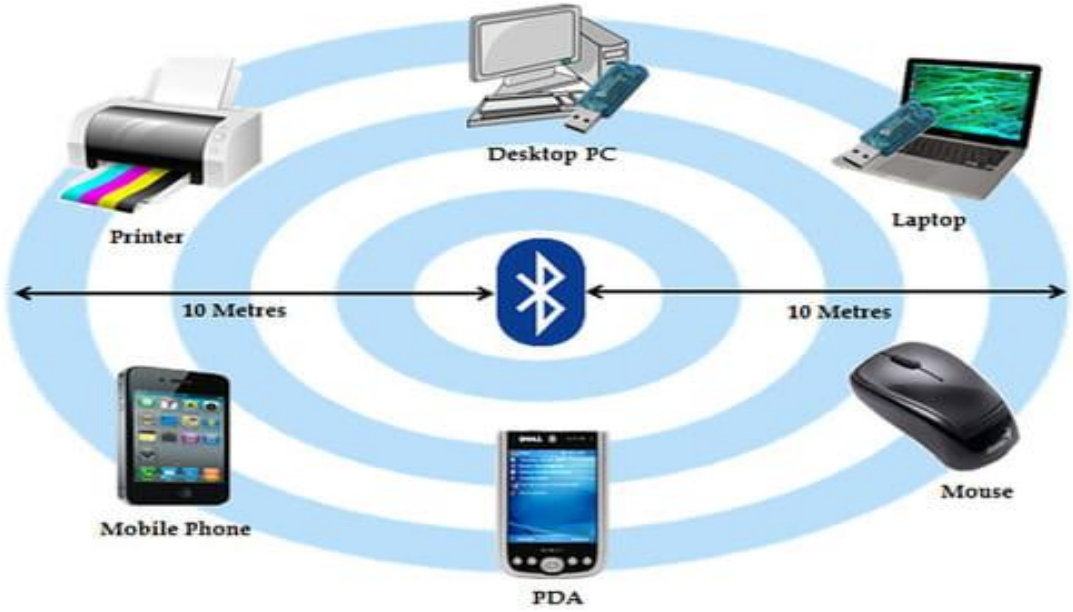
1. Cost: it is less expensive with good data efficiency.
2. High speed: the speed in this network is fast because fiber optics, because MAN uses fiber optics.
3. Sharing of connections: in metropolitan area network it is a combination of two or more LAN networks
4. Security level: it has high security than WAN.

Disadvantages of metropolitan area network:

1. Difficulty in managing: it may be difficult to manage if number of lines are increased. This is due to the security purposes.
2. Speed: speed of network may be slow due to high coverage

Wireless personal area network:

In this network coverage is very limited maximum 100 meters like Bluetooth which help to connect the earpieces to the mobile phones. It uses short range radio waves such as Bluetooth devices such as pointing devices.



Benefits of personal area network:

1. There is no extra space is required.
2. No need of data cables.
3. Data can be transfer between multiple devices at a time.
4. Not much costly.

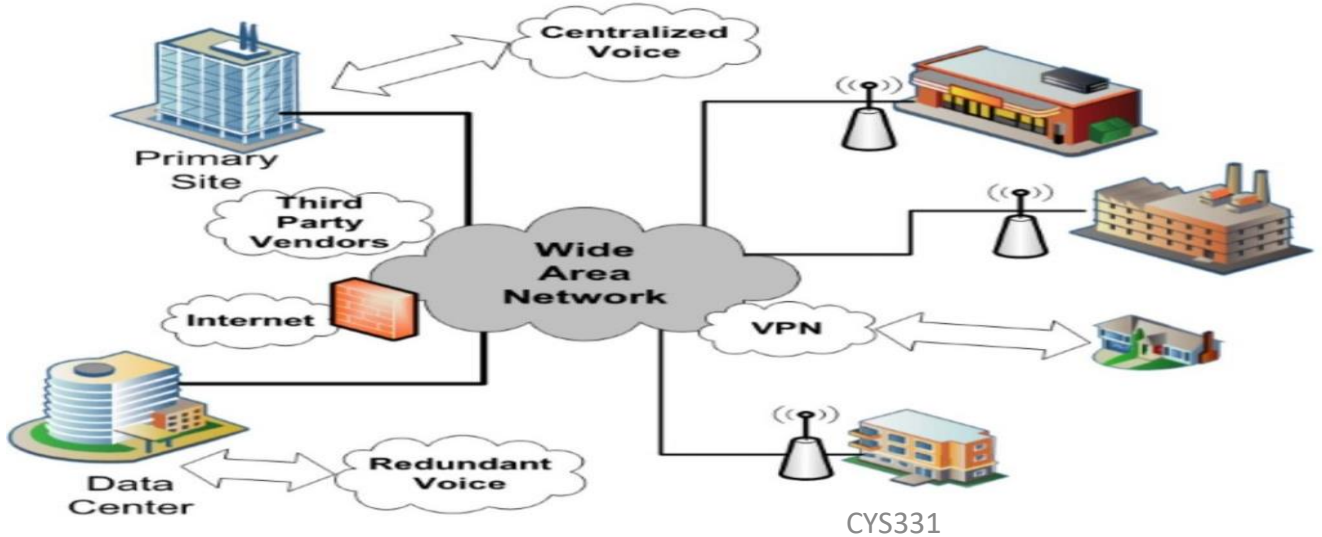
Disadvantages of personal area network:

1. Transferring of data may be slow.
2. It can only be used in personal area limited coverage.
3. It is mostly used in less range of areas

Wireless wide area network

It is different from the rest of the networks because it uses mobile telecommunications and other cellular technologies like 2G, 3G and 4G in transferring of data.

Wireless wide area network allows us to check the mails, browsing using a laptop or connect to VPN within the regional boundaries of cellular service. It basically covers a large geographical area like planes, towers, buildings were included in WWAN.



Thank you